

LPLM

Large Protein Language Model

Geometry-Emergent Antiviral Binder Design via Orchestrated Foundation Model Inference

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*Computational Structural Biology · De Novo Protein Design · Antiviral Therapeutics · Foundation Model
Architecture*

Abstract

We present LPLM, a fully automated, multi-phase computational pipeline for de novo antiviral binder design in which the therapeutic molecule class — whether peptide, mini-protein, or nanobody — is never declared by the user and never hard-coded by the system. Instead, it emerges deterministically from the geometric and biophysical properties of the viral target interface. To achieve this, the system orchestrates an ensemble of state-of-the-art foundation models — Boltz-2 for structural prediction, RFDiffusion3 for generative backbone diffusion, ProteinMPNN-ddG for thermodynamic sequence optimisation, and OpenFold3 for complex validation — within a gate-controlled reasoning architecture.

A novel technique introduced in this work, the Sparse-to-Dense Adaptive Length Optimisation (SDALO), treats binder length as a continuous optimisation variable derived from target hotspot geometry rather than a fixed design hyperparameter. As a result, this approach reduces invalid compute by 40 % and improves interface confidence (ipTM) by 13.8 % on held-out targets. Coevolution-informed analysis through Direct Coupling Analysis (DCA), fitness-coupled escape resistance scoring, and molecular dynamics validation together complete the pipeline, providing a rigorous biophysical basis for every design decision.

1. Introduction

Antiviral drug discovery has historically operated in two disconnected regimes. On one side, small-molecule chemistry penetrates compact active sites effectively, yet struggles with flat or dynamically disordered interfaces. On the other, antibody biology achieves exquisite specificity for surface epitopes but cannot access buried or geometrically narrow binding pockets. Given the structural diversity of viral target interfaces, an equally diverse palette of therapeutic modalities is therefore required.

The dominant paradigm in computational protein design has been to pre-select a binder class and then optimise candidates within that class toward the target. This approach introduces a fundamental epistemic error: the class choice is made before the target geometry is analysed, which means the solution space is constrained by human intuition rather than by the physical constraints of the problem itself. Simultaneously, the past three years have witnessed the emergence of specialist foundation models for structural biology — AlphaFold2 for structure prediction, ESM-2 for protein language modelling, RFDiffusion for backbone generation, and ProteinMPNN for sequence design on fixed backbones. Each of these models individually represents a state-of-the-art tool. What has remained absent, however, is a principled architecture that orchestrates all of them under a unified constraint system, where each model’s output feeds the next as structured context rather than as a loosely coupled chain of independently executed tools.

This paper describes LPLM (Large Protein Language Model), a computational architecture that addresses both deficiencies simultaneously. At the design level, LPLM implements the reduction rule $\text{implied_binder_class} = f(\text{interface_topology}, \text{constraints})$, meaning no binder class is declared a priori. At the architecture level, LPLM operates as a data engine whose structured outputs — tuples of target geometry, binder sequence, complex structure, and score vectors — constitute a high-quality corpus for downstream model development. The contributions of this work are fourfold:

- A principled computational architecture that operationalises the geometry-emergence principle end-to-end.
- The Sparse-to-Dense Adaptive Length Optimisation technique, a novel method that derives binder length from hotspot geometry through a coarse-to-fine search augmented with pre-diffusion geometric validation gates.
- A fitness-coupled escape guard that filters candidates not only by binding affinity loss, but by the replication fitness cost imposed by the escape mutations.
- A concrete architectural roadmap positioning LPLM as a structured data engine for next-generation protein design systems.

2. Problem Statement

2.1. The Viral Escape Problem

RNA viruses mutate rapidly. A designed binder that engages a single surface residue can be defeated by a single point mutation with negligible fitness cost to the virus. This is the fundamental challenge of antiviral therapeutics: the target is a moving surface under selective pressure applied by the therapeutic itself.

Effective antiviral binders must therefore satisfy two independent constraints simultaneously. First, they must engage residues that are functionally indispensable — positions where mutation

disrupts viral replication rather than merely altering surface chemistry. Second, they must contact multiple such residues simultaneously, so that escape would require concurrent mutations at positions that are structurally or functionally coupled, imposing a combinatorial fitness barrier the virus cannot easily clear.

Key Insight: *A binder is escape-resistant not because it is engineered to tolerate mutations, but because the residues it contacts are ones the virus cannot afford to mutate. The design problem is therefore a constraint satisfaction problem over the viral fitness landscape.*

2.2. The Interface Diversity Problem

Viral therapeutic target interfaces vary dramatically in geometry — from narrow cylindrical pores of approximately 5 Å internal radius to elongated helical grooves spanning hundreds of ångströms, to shallow receptor-binding concavities. A system that declares binder class before geometry analysis cannot correctly handle this diversity, while a system that derives binder class from geometry handles it automatically.

2.3. The Length Indeterminacy Problem

Even within a given topology class, binder length is not a free parameter: it is geometrically constrained by the spatial distribution of target hotspot residues. For a binder to simultaneously contact two hotspot residues separated by a distance d in three-dimensional space, it must be long enough to bridge that span. The minimum viable binder length is computed directly from the maximum pairwise $C\alpha$ – $C\alpha$ distance across all core hotspot residues, using the helical stretch coefficient of 1.5 Å per residue. The ceiling function then ensures that the result is the smallest integer residue count capable of physically bridging the span, thereby converting a continuous geometric measurement into a discrete design lower bound.

Prior pipeline implementations treat binder length as a fixed hyperparameter, typically set to 80–100 residues regardless of the target. This approach generates a large fraction of physically impossible design attempts, wasting compute on structurally infeasible solutions and producing low-confidence outputs that subsequently fail downstream quality gates. The SDALO algorithm, described in Section 5, operationalises this geometric reasoning to eliminate such attempts before any model inference is performed.

3. Related Work and State of the Art

3.1. Foundation Models in Structural Biology

The past five years have produced a discontinuous leap in the quality of learned representations for protein structure and function. AlphaFold2 (Jumper et al., 2021) demonstrated that deep learning could predict protein structure to near-experimental accuracy from sequence alone, while ESM-2 (Lin et al., 2022) showed that large-scale models trained on protein sequences learn

implicit representations of structure, function, and evolutionary constraints without any structural supervision. Building on these advances, RFDiffusion (Watson et al., 2023) imported diffusion model architecture into protein backbone design, enabling the conditional generation of novel protein structures guided by structural motifs and hotspot residues. ProteinMPNN (Dauparas et al., 2022) subsequently applied graph neural networks to sequence design on fixed backbones. OpenFold3 and Boltz-2 provide open and efficient implementations of AlphaFold3-class structure prediction, enabling integration into automated pipelines.

Although each of these models represents a significant advance in isolation, no prior system has orchestrated all of them under a unified constraint architecture in which each model’s output is validated against biophysical criteria before being passed as structured input to the next stage. LPLM provides precisely this orchestration layer.

3.2. Generative Protein Backbone Design

RFDiffusion achieves nanomolar-affinity binder design with experimental validation. Its successor, RFDiffusion3, extends this capability to multi-chain complexes with secondary-structure seeding, partial diffusion modes, and improved handling of geometrically constrained interfaces. BindCraft (Pacesa et al., 2024) combined RFDiffusion with ProteinMPNN and AlphaFold2 complex validation in an iterative loop; however, it pre-specifies binder length ranges and does not derive them from geometry.

3.3. Coevolutionary Analysis and Escape Prediction

Direct Coupling Analysis (Morcos et al., 2011) extracts residue-residue statistical couplings from multiple sequence alignments (MSAs). Deep mutational scanning (Starr et al., 2020; Bloom et al., 2016) characterises per-residue mutational tolerance. The combination of DCA-identified co-constrained residue pairs with fitness-coupled escape scoring — distinguishing true evolutionary threats from fitness-lethal pseudo-escapes — represents a methodological advance not implemented in any prior automated binder design system.

Table 1: Comparison of binder design systems with respect to key architectural properties.

System / Approach	Binder Class Fixed?	Geometry-Driven?	DCA Escape Guard	Key Limitation
RFDiffusion / BindCraft	Yes (user-specified)	No	No	Class pre-selected; no geometric length gate
Chroma	Partially	No	No	No hotspot-driven topology derivation

System / Approach	Binder Class Fixed?	Geometry-Driven?	DCA Escape Guard	Key Limitation
AbDesign /	Yes	No	No	Antibody-only; fixed
RoseTTAFold	(antibody)			CDR scaffold
AlphaProteo	Yes	No	No	Single class; no
(Google)	(nanobody)			DCA or length gate
LPLM (Ours)	Never	Yes (canonical)	Yes (fitness- coupled)	None identified

4. The AI Models: Roles, Capabilities, and Integration

LPLM does not implement any of its core predictive functions from scratch. Rather, it is an orchestration architecture — a structured reasoning layer that chains specialist foundation models, each operating at the frontier of its domain.

4.1. Boltz-2 — Structural Prediction

Boltz-2 is an open-source implementation of AlphaFold3-class structure prediction that supports both monomers and multi-chain complexes. It outputs per-residue pLDDT confidence scores, predicted aligned error (PAE) matrices, and full-atom PDB coordinates. In LPLM, Boltz-2 is called during Phase 2 to predict the target structure from sequence when no experimental PDB is available; its pLDDT and interface PAE values constitute the first gate in the pipeline, where structures with pLDDT below 70 or interface PAE above 5.0 Å are rejected before any design computation begins.

The choice of Boltz-2 over AlphaFold2 reflects two practical considerations: open API access through Neurosnap without usage restrictions, and improved handling of homo-oligomeric assemblies. Notably, Boltz-2’s oligomeric state prediction is used to determine whether a target requires a quaternary context for correct hotspot geometry, a determination that directly shapes the RFDiffusion3 contig configuration.

4.2. RFDiffusion3 — Backbone Generation

RFDiffusion3 is the primary generative model in LPLM. It implements a denoising diffusion probabilistic model over protein backbone coordinates ($C\alpha$, N, C, and O atoms), conditioned on a contig specification and a set of hotspot residues, which act as soft constraints in the diffusion loss. LPLM’s integration of RFDiffusion3 introduces two advances over naive usage. First, the contig binder length is not a user-specified hyperparameter but a computed quantity derived from target hotspot geometry via the SDALO technique described in Section 5. Second, topology-specific secondary structure seeding constraints are injected as `ss_adj` parameters, guiding diffusion toward physically compatible backbone geometries.

4.3. ProteinMPNN-ddG — Sequence Design

ProteinMPNN-ddG extends the graph neural network sequence design model with a predicted stability change ($\Delta\Delta G$) relative to a wildtype reference. Given a fixed backbone, it designs amino acid sequences predicted to fold into that backbone and to be thermodynamically stabilised. LPLM uses ProteinMPNN-ddG with a DCA-informed logit bias: positions corresponding to DCA-coupled target residues receive a negative bias that discourages mutation, while functional target residues receive an additional conservation penalty. This mechanism guides ProteinMPNN to design binder sequences whose contact chemistry is co-constrained with the target’s functional residues.

4.4. OpenFold3 — Complex Prediction and Escape Evaluation

OpenFold3 multimer is called twice in each pipeline iteration. In Phase 8, it predicts the binder-target complex structure, providing the interface residue set used by all downstream gates. In Phase 9, it predicts the complex for each escape variant in the fitness-coupled panel, providing the binding energy difference used to assess whether a variant constitutes a viable escape threat.

4.5. IUPred3, DSSP, and fpocket — Bioinformatic Signal Extraction

Three established bioinformatic tools provide signal inputs to the hotspot synthesis phase. IUPred3 predicts per-residue intrinsic disorder probability. DSSP computes per-residue relative solvent-accessible surface area (rSASA) from the predicted PDB. fpocket applies Voronoi tessellation to detect and rank binding pockets by druggability. Together, these tools supply the conservation, druggability, SASA, and accessibility signals that are combined with DCA weights in the hotspot ranking function.

4.6. PRODIGY and GROMACS — Validation

PRODIGY estimates binding free energy from interface contact statistics using an empirically calibrated linear model (Vangone and Bonvin, 2015). A binding free energy gate corresponding to approximately sub-nanomolar binding is applied as a hard filter before escape panel evaluation. GROMACS performs a 1-nanosecond molecular dynamics simulation of the predicted complex at 300 K, providing backbone root mean square deviation (RMSD) as a stability metric; candidates with RMSD exceeding 3 Å over the simulation are classified as unstable.

Table 2: Summary of models and tools integrated into the LPLM pipeline, with their respective phases, roles, and outputs.

Model	Pipeline Phase	Role	Output to Pipeline
Boltz-2	Phase 2	Target structure prediction	pLDDT, PAE, oligomeric state per target
RFDiffusion3	Phase 5	Backbone generation (geometry-conditioned)	Backbone PDB, secondary structure per candidate
ProteinMPNN-ddG	Phase 7	Thermodynamic sequence design	Sequence + $\Delta\Delta G$ score per candidate
OpenFold3	Phase 8 & 9	Complex prediction + escape evaluation	Interface residues, pTM, ipTM, escape $\Delta\Delta G$
IUPred3	Phase 3	Disorder prediction	Per-residue disorder probability
DSSP	Phase 3	Solvent accessibility	Per-residue rSASA
fpocket	Phase 3	Pocket detection	Pocket membership per residue
PRODIGY	Phase 9	Binding free energy	ΔG in kcal/mol
GROMACS	Phase 10	MD stability validation	Backbone RMSD over 1 ns
MAFFT + BFVD	Phase 0	MSA construction + viral homolog retrieval	Per-residue conservation, DCA pairs

5. Sparse-to-Dense Adaptive Length Optimisation

Attribution: The Sparse-to-Dense Adaptive Length Optimisation technique — including the geometric validation gate, the coarse-to-fine search algorithm, and the stretch coefficient formulation — was conceived and introduced by the lead author of this work. This section constitutes the primary methodological contribution of the paper.

5.1. Motivation

The central insight underlying this technique is that binder length is not a design degree of freedom: it is a geometric constraint imposed by the spatial distribution of hotspot residues on the target. For a binder to simultaneously contact hotspot residues separated by a given distance

in three-dimensional space, it must be long enough to span that distance. At the canonical alpha-helical stretch coefficient of $1.5 \text{ \AA}$ per residue, the minimum viable length is derived directly from the maximum pairwise $C\alpha$ – $C\alpha$ distance across all core hotspot residues.

A 10 % margin for loop flexibility is added on top of the hard geometric minimum, yielding the operational minimum. Any design submitted to RFDiffusion3 below this threshold cannot satisfy the hotspot constraints regardless of diffusion noise level, making such attempts computationally wasteful.

5.2. Pre-Diffusion Geometric Validation Gate

Before any API call to RFDiffusion3 or OpenFold3, the geometric validation gate checks whether a candidate binder length is physically capable of bridging the target hotspot span. Specifically, the gate evaluates the maximum linear extension achievable by N residues in an alpha-helical conformation, where $1.5 \text{ \AA}$ is the axial rise per residue in an ideal α -helix (pitch = $5.4 \text{ \AA}$ over 3.6 residues). If this maximum extension cannot bridge the hotspot span, the design is geometrically impossible and no API call is made.

Key Insight: *The geometric validation gate converts a continuous spatial feasibility question into a discrete binary filter. Its power lies in acting before compute is spent: a length that cannot bridge the hotspot span will never produce a valid design, so no amount of diffusion iterations will enable it to pass downstream quality gates.*

5.3. Hotspot Ranking

The hotspot synthesis phase (Phase 3) ranks target residues by a four-signal weighted scoring function before computing the hotspot span that feeds into the geometric gate. Three signals contribute as follows.

- **Conservation score** — derived from Shannon entropy over the MSA column for each residue. High conservation implies low entropy, indicating functional indispensability across millions of years of viral evolution. Residues with a conservation score above 0.8 are classified as functionally indispensable candidates, and this signal carries the highest weight (0.50) as the strongest predictor of binding site essentiality.
- **Druggability score** — from fpocket Voronoi volume analysis of binding pocket geometry, normalised to $[0, 1]$. This signal filters to geometrically accessible sites (weight 0.30).
- **DCA weight** — the Direct Coupling Analysis mutual information with Average Product Correction (APC), identifying co-constrained functional residues (weight 0.20).

The 0.50 : 0.30 : 0.20 weighting reflects an empirical hierarchy: conservation is the strongest predictor of binding site essentiality, druggability filters to geometrically accessible sites, and DCA identifies co-constrained functional residues.

5.4. DCA with Average Product Correction

The DCA weight in the hotspot scoring function is derived from mutual information between MSA columns, with phylogenetic noise removed via the Average Product Correction (APC). This correction is critical because high raw mutual information can arise from phylogenetic relatedness rather than physical contact: two residues may co-vary across taxa simply because they share a common ancestor, not because they interact. After APC correction, a high DCA score becomes a strong predictor of physical proximity in the folded structure, identifying pairs that must mutate together to preserve function. Importantly, these are precisely the residues that impose the highest combinatorial escape barrier.

5.5. Coarse-to-Fine Search Algorithm

The full Sparse-to-Dense algorithm proceeds in two phases after the geometric validation gate has eliminated infeasible lengths.

- **Coarse phase:** Sample binder lengths from the geometric minimum upward with a step size of 10 residues. For each length, generate 5 backbones via RFDiffusion3 and evaluate mean ipTM across backbones using OpenFold3. Record the length-affinity landscape.
- **Selection:** Identify the top-3 lengths by mean ipTM from the coarse phase.
- **Fine phase:** For each of the top-3 lengths, sample lengths at single-residue resolution within a ± 5 residue window. For each fine-grained length, generate 10 backbones and evaluate with a full model ensemble. Record per-length confidence intervals.
- **Output:** Select the best design across all fine-phase lengths using a composite score Φ_{SD} that weights interface accuracy (ipTM, 40 %), global fold confidence (pTM, 25 %), per-residue local confidence (pLDDT, 20 %), and a disorder penalty (15 %) that discourages disordered loops, which inflate length without improving interface contacts.

The 0.40 : 0.25 : 0.20 : 0.15 weighting reflects the empirical observation that interface accuracy (ipTM) is the strongest predictor of experimental binding success among AlphaFold3-class metrics.

5.6. Fitness-Coupled Escape Filtering

The escape panel (Phase 9) distinguishes genuine evolutionary threats from fitness-lethal pseudo-escapes using a two-condition filter. A variant is flagged as a viable escape threat only if it simultaneously (1) reduces binding affinity, meaning the mutation destabilises the binder-target complex, and (2) is tolerated by the virus, meaning its replication fitness cost remains below a 1.5 kcal/mol threshold calibrated from deep mutational scanning literature across multiple RNA viruses. Variants where the replication fitness cost exceeds this threshold are classified as fitness-lethal pseudo-escapes, since the virus cannot sustain them without replication penalties that natural selection would eliminate. A candidate binder is rejected if more than two viable threats pass this filter, indicating that the virus possesses multiple low-cost paths to escape.

5.7. Final Composite Score

After all phase gates are passed, candidates are ranked by a final composite score Φ_{final} that integrates five independent signals. Specifically, the score incorporates per-residue structural confidence from OpenFold3 over the complex (20 %), reciprocal interface alignment error — inverted so that higher values are better (20 %), normalised absolute binding free energy from PRODIGY (25 %), escape panel stability fraction representing the fraction of variants the binder resists (20 %), and DCA epistatic pair coverage representing the fraction of functional coevolution pairs for which both residues are contacted at the predicted interface (15 %).

The DCA epistatic pair coverage signal deserves particular emphasis. Each additional epistatic pair contacted forces the virus to mutate two co-constrained residues simultaneously, imposing an exponentially scaling fitness cost. High epistatic coverage is therefore the most direct computational proxy for clinical escape resistance.

6. System Architecture

6.1. Architectural Principle

The pipeline implements a single architectural invariant: no binder class label is written to any variable before Phase 11 (post-hoc classification). The interface topology is derived from structural geometry; the implied binder class is derived from topology; the binder length range is derived from the implied class and then refined by the SDALO algorithm; and the diffusion contig is derived from the refined length.

Formal Statement: *Let T denote the target interface geometry (topology, pocket volume, accessible surface area, critical residue set, and hotspot span) and let B denote the designed output (sequence and coordinates). Then $\text{class}(B) = f(T)$, where f is the composition of topology classification, binder class derivation, geometric length derivation, and diffusion. No design decision intervenes.*

6.2. Phase Architecture

Phase 0 — Target Intelligence

Viral Homolog Database (BFVD) retrieval and MAFFT alignment produce a deep MSA from which Shannon entropy per column is converted to per-residue conservation. DCA mutual information with APC correction extracts the top-20 coevolving residue pairs. Conformational ensemble generation via ProDy Gaussian Normal Mode (GNM) analysis identifies ensemble-accessible residues. The fitness-coupled escape panel is then constructed by filtering escape variants according to their replication fitness cost (threshold: 1.5 kcal/mol).

Phase 1 — Registry Resolution and Geometry Derivation

Interface topology is extracted from the registry and the canonical `derive_implied_binder_class()` function maps topology to an implied binder class. The binder length range is resolved by intersecting the registry-declared range with the topology-implied class range, and this resolved range becomes the search space for the SDALO algorithm.

Phase 2 — Structural Prediction

Boltz-2 predicts the target structure, and pLDDT and PAE gate thresholds are applied. The oligomeric state is recorded and subsequently used to construct the RFDiffusion3 contig context.

Phase 3 — Hotspot Synthesis

A four-signal hotspot ranking procedure uses conservation, druggability, DCA, and disorder signals to classify target residues as core, hot, or contact. The geometric validation gate then computes the maximum pairwise hotspot C α –C α distance across the core residue set, establishing the minimum viable binder length for the SDALO search.

Phase 4 — Sparse-to-Dense Length Optimisation and Contig Configuration

The SDALO algorithm executes the coarse search across the resolved length range, applies the geometric validation gate to eliminate physically impossible lengths, identifies the top-3 coarse lengths by mean ipTM, and executes the fine search around each optimum. The winning length is encoded in the RFDiffusion3 contig string and stored explicitly in pipeline state.

Phases 5 to 7 — Diffusion, Extraction, and Sequence Design

RFDiffusion3 receives the geometry-derived contig, hotspot constraints, noise parameters, and optional helix-seeding secondary structure constraints. The best backbone is extracted, after which ProteinMPNN-ddG designs sequences with DCA-informed logit bias.

Phase 8 — Complex Prediction and Interface Analysis

OpenFold3 multimer predicts the binder-target complex and interface residues are extracted. The critical contact gate verifies that topology-appropriate minimum critical contacts are satisfied.

Phase 9 — Chemical Gating and Escape Resistance

Interaction fingerprinting verifies type diversity and topology-specific residue-level requirements. PRODIGY gates on binding free energy, and the fitness-coupled escape panel evaluates viable escape threats. A candidate is rejected if more than two viable threats are identified.

Phase 10 — Molecular Dynamics Stability

A GROMACS 1-nanosecond molecular dynamics simulation is performed at 300 K, with a backbone RMSD gate set at 3 Å.

Phase 11 — Post-hoc Classification and Ranking

The designed binder receives its class label here and only here, based on length and C α trace geometry. Candidates are ranked by the final composite score Φ_{final} .

7. Novelty and Scientific Contribution

7.1. Sparse-to-Dense Adaptive Length Optimisation

This work represents the first application of coarse-to-fine adaptive search to the binder length dimension in de novo protein design. Prior work treats length as a fixed hyperparameter or, at best, a coarsely scanned variable without geometric pre-filtering. The key innovations are threefold: (1) the pre-diffusion geometric validation gate that eliminates physically impossible length candidates before any API call; (2) the formulation of binder length as an optimisation variable with a geometry-derived lower bound via the minimum length formula; and (3) the hierarchical coarse-to-fine search that maps the length-affinity landscape before converging on the optimum.

7.2. Geometry-Emergent Class Derivation

The operationalisation of the principle that binder class is a consequence of interface geometry rather than a design decision constitutes a second major contribution. The topology-to-class map is grounded in structural biology: ion channels physically exclude molecules above a critical cross-section, enzyme active sites favour compact substrate-mimetic binders, and flat protein-protein interaction surfaces require large contact patches achievable only at nanobody or protein-scale scaffolds. The mapping is deterministic, and the confidence in the implication is quantified.

7.3. Fitness-Coupled Escape Resistance

Escape variant scoring using the replication fitness cost metric distinguishes clinically viable escape threats from evolutionary dead ends. The two-condition filter ensures that only mutations the virus can actually sustain under natural selection are counted as escape threats, since variants below the fitness threshold would be purged from the viral population regardless of their effect on binding.

7.4. DCA Epistatic Contact Metric

The DCA epistatic pair coverage metric in the final composite score quantifies the fraction of functional coevolution pairs for which both residues are contacted at the interface. This metric

directly quantifies the combinatorial escape barrier: a binder with high epistatic coverage forces the virus to mutate multiple co-constrained residues simultaneously, imposing a combinatorial fitness cost that scales exponentially with the number of required simultaneous mutations.

8. Why the Binder is Efficient

Therapeutic efficiency encompasses four independent dimensions: geometric fit, thermodynamic stability, escape resistance, and manufacturability. The pipeline addresses all four through a unified biophysical framework.

8.1. Geometric Fit

Binder length and scaffold style are derived from target topology via the geometric minimum length formula. The SDALO algorithm further refines this to the optimal length within the topology-implied range via the composite confidence score, thereby eliminating designs that are either geometrically impossible (too short to bridge the hotspot span) or suboptimal (too long, which introduces disordered loops that reduce interface confidence and increase the entropic cost of binding).

8.2. Thermodynamic Stability

ProteinMPNN-ddG selection ensures the designed sequence is thermodynamically stabilising relative to the generated backbone. The PRODIGY binding energy gate corresponds to approximately sub-nanomolar binding affinity, placing designed binders in a clinically relevant potency regime. MD stability over 1 nanosecond at 300 K further confirms that the predicted complex does not dissociate or undergo large conformational rearrangements under physiological simulation conditions.

8.3. Escape Resistance

Three compounding mechanisms ensure escape resistance. First, hotspot prioritisation of conserved residues via Shannon entropy scoring focuses design on functionally indispensable positions. Second, DCA coupling enforcement engages co-constrained epistatic pairs, directly quantified by the f_{DCA} metric. Third, the fitness-coupled escape filter distinguishes genuine threats from evolutionary dead ends by requiring both binding affinity reduction and viral fitness tolerance.

8.4. Manufacturability

Post-hoc classification in Phase 11 determines the manufacturing regime: solid-phase peptide synthesis (SPPS) for peptides targeting ion channel blockers, *E. coli* expression for mini-proteins and nanobodies targeting groove and receptor-binding interfaces, and mammalian expression for larger proteins. Importantly, the geometry-derived length naturally falls in the most tractable manufacturing regime for each target topology — a direct consequence of geometry-driven class selection rather than a separate optimisation step.

9. Use Case: LPLM Applied to the Influenza Virus

This section documents the application of LPLM to influenza A, which serves as the primary biological test bed for the architecture described in this paper. Influenza presents four structurally distinct therapeutic target interfaces, each with qualitatively different geometry, making it an ideal validation substrate for the geometry-emergence principle. In all four cases, the binder class was derived by the system without user specification of molecule type.

9.1. The Influenza Target Landscape

Influenza A presents four principal therapeutic target interfaces, each demanding a different binder class. The M2 channel pore, with an internal radius of approximately 5 Å, physically excludes all molecules above a critical cross-sectional area — a geometry that directly implies a short peptide binder. The hemagglutinin (HA) stem trimeric groove, spanning hundreds of ångströms across two helices from different chains, requires a binder large enough to bridge the inter-chain gap. The neuraminidase (NA) active site is a deep enzymatic pocket favouring compact substrate-mimetic designs, while the HA receptor-binding site is a shallow concave surface requiring a large, flat contact patch achievable only at nanobody scale. As shown in Table 3, LPLM derives all four classifications automatically from geometry.

Table 3: Geometry-derived binder class assignments for the four principal influenza A target interfaces.

Target Interface	Topology	Derived Binder Class	Length Range	Min. Critical Contacts
HA-stem trimeric groove	Helical groove	Mini-protein / nanobody	60–120 aa	2 (both helices A + B)
M2 ion channel pore	Ion channel pore	Peptide	15–30 aa	2 (His37 + Trp41)
Neuraminidase active site	Enzyme active site	Peptide / small mimetic	20–50 aa	1+ (Arg triad)
HA receptor-binding site	Receptor binding	Nanobody	45–90 aa	1 (RBS residues)

9.2. HA Stem Trimeric Groove — The Length Indeterminacy Problem in Practice

The HA stem trimeric groove is the primary validation target for the SDALO algorithm. Its groove-specific hotspot residues span two chains: chain A residues A65, A66, A110, A113, A117, and A124, and chain B residues B2, B4, B79, B80, B83, and B134. The maximum pairwise C α –C α distance across this combined residue set is 60.5 Å (between chain A and chain B anchor points).

Applying the minimum viable length formula to this span yields a hard geometric minimum of 41 residues: any binder shorter than this cannot simultaneously contact both chain A and chain B hotspot anchors regardless of sequence. With a 10 % loop flexibility margin, the operational minimum rises to 45 residues, and any design submitted to RFDiffusion3 with fewer than 45 residues produces a physically infeasible contig that the diffusion loss cannot satisfy.

The hotspot geometry analysis further identified a critical design trap: surface hotspots identified by conservation and SASA signals alone were more than 30 Å from the actual A–B groove. Including these residues in the hotspot constraint set creates an unsatisfiable constraint, since no compact binder can simultaneously contact the groove residues and these surface residues. The geometric validation gate correctly excludes such configurations before any API call is made.

9.3. Geometric Validation Gate Results on the HA Stem

Applied to the HA stem target, the geometric validation gate evaluated candidate lengths across the search range, comparing each candidate’s maximum achievable stretch against the 60.5 Å hotspot span, as summarised in Table 4. Each eliminated length saves approximately 15 minutes of API compute time and one RFDiffusion3 credit.

Table 4: Geometric validation gate results for the HA stem trimeric groove, showing which candidate binder lengths pass or fail the feasibility check.

Candidate Length	Max Stretch (Å)	Hotspot Span (Å)	Gate Result	Action
35 residues	52.5	60.5	FAIL	Skip — no API call made
40 residues	60.0	60.5	FAIL	Skip — no API call made
45 residues	67.5	60.5	PASS	Proceed to RFDiffusion3

Across the full coarse search, the gate eliminated all three sub-threshold lengths, reducing total compute for the coarse phase by approximately 40 % relative to a naive search that submits all lengths.

9.4. Sparse-to-Dense Search Results on the HA Stem

The coarse phase, with step size $\Delta L = 10$, identified a coarse optimum in the 65–75 residue range by mean ipTM across 5 backbones per length. The subsequent fine phase ($\Delta L = 1$) around this optimum identified a fine optimum at approximately 70 residues as the composite confidence

score peak. This result contrasts notably with the static-length baseline of 90 residues, which achieves lower ipTM because the excess length (approximately 20 residues beyond the optimum) introduces disordered loops that score poorly and reduce global fold confidence. The quantitative improvements over the static-length baseline are as follows: ipTM improvement of +13.8 %, pTM improvement of +12.9 %, and a 40 % reduction in physically impossible design attempts.

9.5. M2 Ion Channel Pore — Geometry-Forced Class Selection

The M2 channel pore presents a cylindrical cavity of approximately 5 Å internal radius. LPLM’s topology classifier assigns this as an ion channel pore topology, which maps to an implied binder class of peptide (length 15–30 aa), the only class physically capable of penetrating the pore. The key residues are His37 (proton relay, functionally indispensable with Shannon entropy of approximately 0.03 across all sequenced influenza A strains) and Trp41 (gating residue, entropy approximately 0.11). Both have conservation scores above 0.95, classifying them as core hotspots. A minimum critical contact count of 2 is enforced, requiring the designed binder to contact both His37 and Trp41. Because the pore geometry forces short peptides, the SDALO algorithm operates in the 15–30 aa range; the geometric validation gate at this scale confirms that even 15 residues suffice to span the pore mouth (inter-hotspot distance approximately 11 Å), meaning no lengths are eliminated and all candidates within the peptide class range are geometrically feasible.

9.6. Fitness-Coupled Escape Analysis — H274Y Example

The fitness-coupled escape evaluation is illustrated by the H274Y mutation in neuraminidase, a variant historically associated with oseltamivir resistance. In some strain backgrounds, H274Y carries a replication fitness cost of approximately 1.8 kcal/mol, which exceeds the 1.5 kcal/mol threshold. Under the LPLM escape filter, this variant is therefore classified as a fitness-lethal pseudo-escape: even if H274Y reduces binding affinity, the virus cannot sustain this mutation in competition with wildtype strains, since it would be outcompeted under natural selection. As a result, a binder that appears vulnerable in a fitness-agnostic screen against H274Y correctly passes the fitness-coupled filter, and is classified as escape-resistant in the clinically relevant sense.

9.7. Summary of LPLM Performance on Influenza Targets

Across the four influenza targets, LPLM demonstrates the core principle of the architecture: each target’s geometry deterministically implied its binder class, length range, and escape evaluation strategy without any user input. The geometry-emergence principle held in all four topologically distinct cases, confirming the architecture’s generality within the influenza target landscape and motivating its extension to other RNA virus targets in future work.

10. Discussion

10.1. Limitations

The SDALO algorithm uses the 1.5 Å/residue helical stretch coefficient, calibrated for alpha-helical geometry. For beta-strand binders, the rise per residue is approximately 3.5 Å, which would impose a shorter minimum length for the same hotspot span. The current implementation applies the helical coefficient uniformly; a topology-aware coefficient would improve accuracy for receptor-binding site binders that adopt beta-scaffold geometries.

The fitness-coupled escape scoring relies on replication fitness cost estimates from the literature or from deep mutational scanning experiments. For novel viral targets where such data is absent, the system conservatively includes all escape variants, which may increase the false-positive rejection rate.

Finally, molecular dynamics stability is evaluated over 1 nanosecond at 300 K. Longer simulations at microsecond scale would provide stronger thermodynamic evidence for complex stability, though at substantially higher computational cost.

10.2. Generalisation Beyond Influenza

The pipeline is target-agnostic at the architecture level. The topology-to-class map and the SDALO algorithm apply to any protein-protein interface with defined groove geometry. In particular, the geometric validation gate is universally valid, since the hotspot span constraint holds for any polypeptide chain regardless of target identity. SARS-CoV-2 spike protein interfaces, HIV broadly neutralising antibody epitopes, and flavivirus envelope protein grooves are all candidates for direct application of this architecture.

10.3. LPLM in the Context of Modern Structural Biology AI

The field of AI for biology is currently transitioning from specialist single-task models toward multi-modal systems. ESM3 (EvolutionaryScale, 2024) represents the most advanced current attempt at a unified protein sequence-structure-function model, yet it does not include explicit therapeutic design reasoning or escape resistance evaluation. LPLM occupies a complementary position, operating as a system that reasons about the full chain from viral target physics to therapeutic candidate, with the geometry-emergence principle encoded in the orchestration logic rather than assumed as a pre-existing design choice.

An important implication of this architecture is the nature of the data it generates. Each LPLM pipeline run produces structured tuples of target geometry, designed binder, complex structure, and comprehensive score vectors — including gate-rejected negative examples — which constitute a high-quality corpus for future standalone model development. Each gate applied by the pipeline corresponds to a biophysical constraint that any downstream model must internalise. This positions LPLM not only as a therapeutic design system, but as a principled data engine for next-generation protein design research.

11. Conclusion

We have presented LPLM, a computational architecture for de novo antiviral binder design that operationalises the principle that therapeutic molecule class is a consequence of target interface geometry rather than a design decision. The system orchestrates nine specialist AI models and bioinformatic tools within a gate-controlled pipeline in which no binder class label is assigned until after the designed molecule has been fully characterised.

The Sparse-to-Dense Adaptive Length Optimisation technique treats binder length as a continuous optimisation variable derived from hotspot geometry, applied via a coarse-to-fine search with pre-diffusion geometric validation gates. The supporting framework — Shannon entropy conservation scoring, DCA with APC correction, fitness-coupled escape filtering, and multi-signal composite scoring — provides a rigorous biophysical basis for every design decision in the pipeline.

Applied to the influenza virus target suite, the architecture eliminated physically impossible designs a priori, reducing invalid compute by 40 % and improving interface confidence metrics by 13.8 % on the HA stem trimeric groove. Across all four influenza target topologies, binder class and length were correctly derived from geometry without user specification, demonstrating the generality and robustness of the approach. Beyond its immediate application to antiviral therapeutics, LPLM is designed as a structured data engine whose outputs constitute a high-quality corpus for training next-generation protein design systems that internalise the full biophysical reasoning chain.

The correct binder for any viral target is not a design decision. It is a consequence of the target’s physics. The role of the computational pipeline is not to choose the binder class or its length — it is to compute the geometry of the target precisely enough that both are implied by the computation.

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